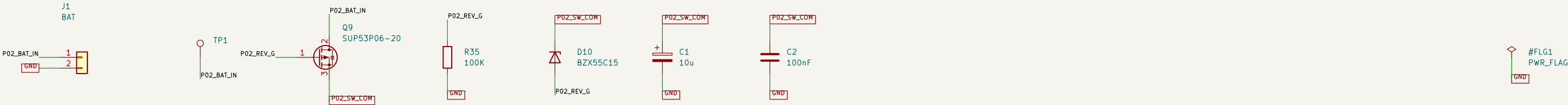
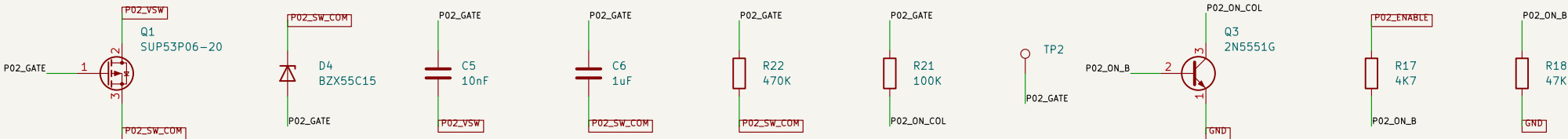


P02-R4 / 01 WEJ: WEJSCIE PAKIETU 4S, Q9/Q1/Q2, VSW, VMOTOR, PODTRZYMANIE C_H

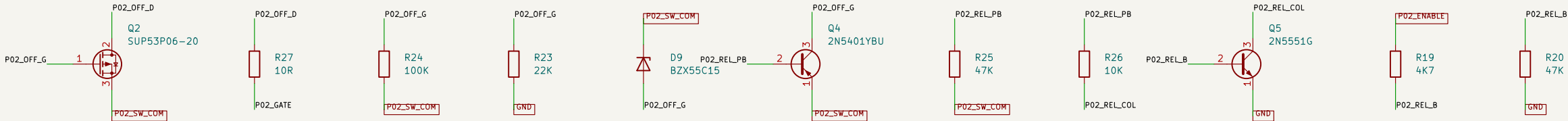
Pakiet 4S -> J1 -> Q9 (Q_REV: ochrona polaryzacji, zrodla Q9/Q1 w SW_COM)



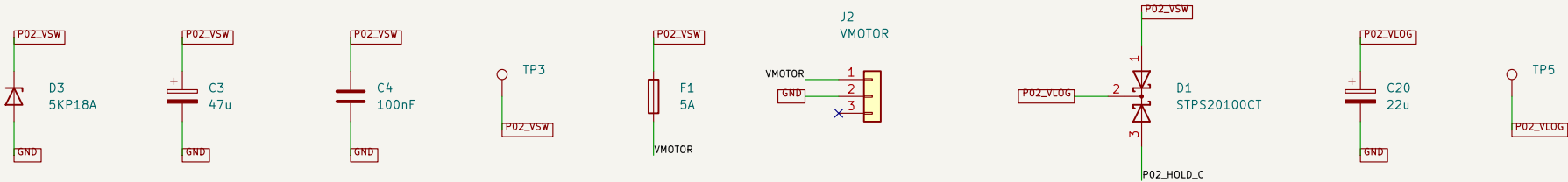
Q1 (Q_SW) - włącznik; załączanie przez Q3/R21, narastanie VSW ustala C5 (jak P01 R3)



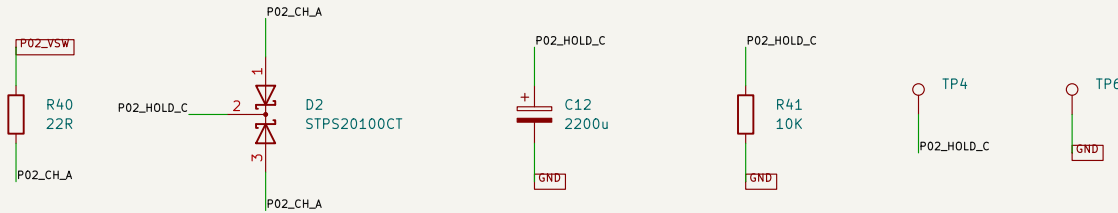
Q2 (Q_OFF) - szybkie wylaczenie; Q4/Q5 zwalniaja Q2 przy ENABLE. R23 = 22k / 0,5W (decyzja 29.09)



VSW: TVS, C, VMOTOR przez F1 5A; D1a/D1b -> VLOG



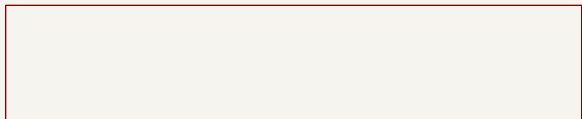
Podtrzymanie: R40 22R bezpiecznikowy -> D2 -> C_H 2200u (-> D1b -> VLOG)



Z-06: pojemnosc zalaczana przez Q1 <= 220 uF laczenie: C3 47u + VLOG (C20 22u, C21/C23 10u) + wejscie P07 <= 131 uF.

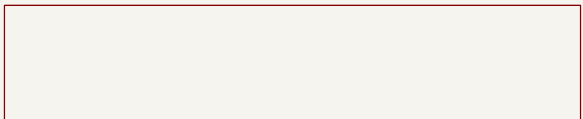
C_H ladowany osobno przez R40 + D2 (nie liczy sie do 220 uF). Hold-up po PFAIL_N: docs/OBLICZENIA-R4.md.

STER



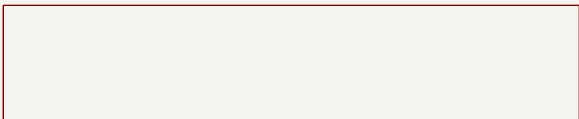
Plik: STER.kicad_sch

LV



Plik: LV.kicad_sch

MON



Plik: MON.kicad_sch

Tor bramki Q1/Q2 i sterowanie z P01 R3 (R23 = 22k / 0,5W)
Zasilanie z pakietu Li-ion 4S (12,0-16,8 V); UVLO 13,53 / 12,51 V; C_H 2200 uF
EGRLab / schematic review (etap 1, bez PCB)

Sheet: /
File: P02.kicad_sch

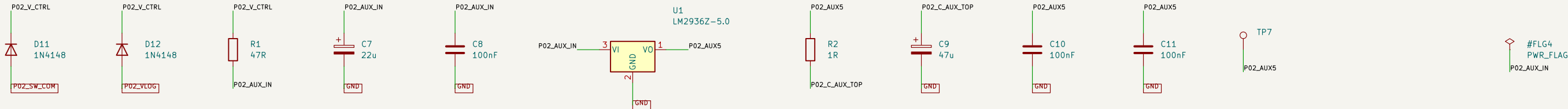
Title: P02-R4 / WEJ

Size: A3 Date: 2026-09-29 Rev: P02-R4

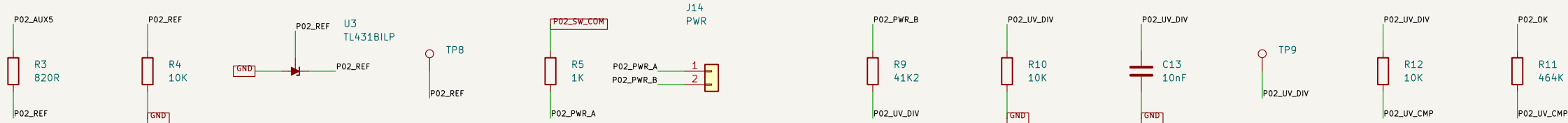
KiCad E.D.A. 10.0.6 Id: 1/4

P02-R4 / 02 STER: STEROWANIE: AUX5, REF, UVLO, OK, ENABLE, PFAIL_N, SAFE_N/PG, LED

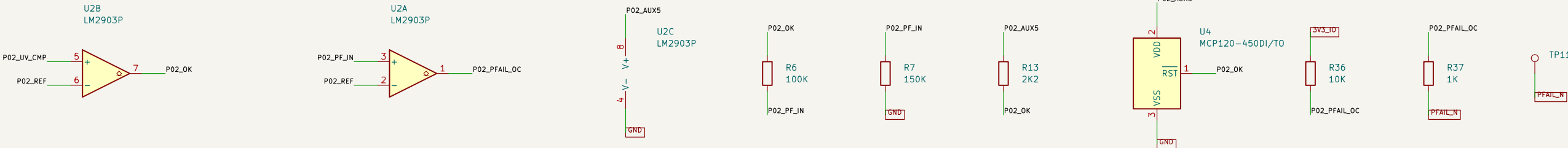
AUX5: LM2936 zasilany z SW_COM lub VLOG (D11/D12) – trwa przez podtrzymanie



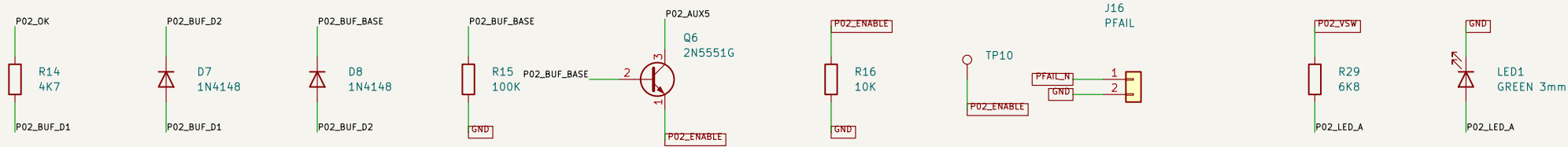
REF 2,495 V; UVLO: SW_COM – R5 – PWR (J14) – R9 – UV_DIV (R10, C13) – R12 – UV_CMP <- R11 <- OK



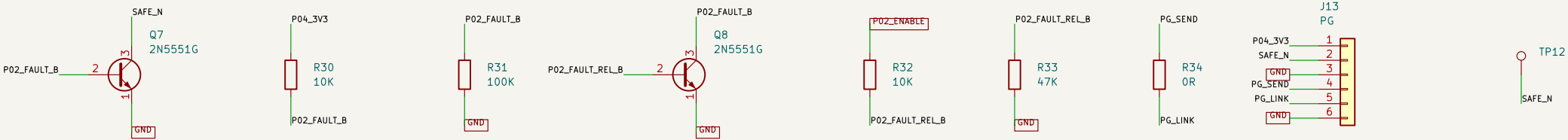
U2B: UVLO -> OK (histereza R11). U2A: PFAIL_N = bufor OK (0,6 x OK wobec REF) -> R37 -> J16.1, J12.3



ENABLE = OK przez D7/D8 i wtornik Q6 (jak P01 R3); LED PWR na VSW



SAFE_N i PG do P04 (z P01 R3): Q7 zasilany z P04_3V3 (J13.1), Q8 zwalnia przy ENABLE; R34 zwora PG_SEND-PG_LINK

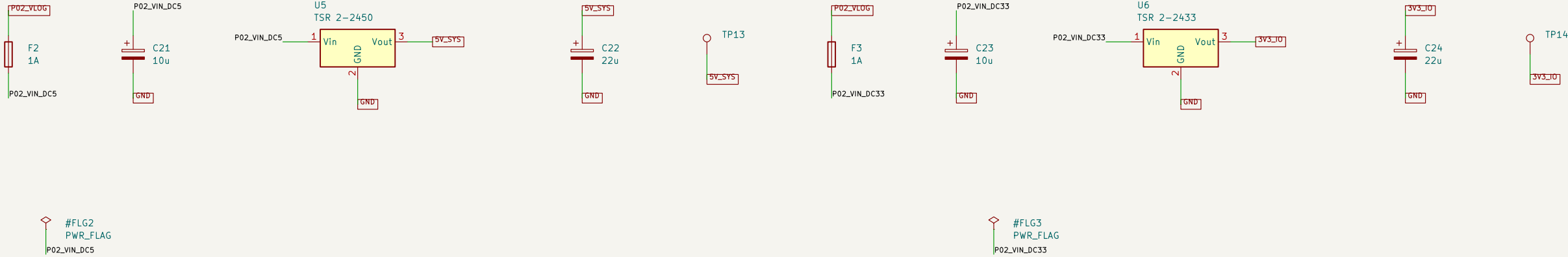


PWR rozwarty albo przerwany przewod = UV_DIV do masy = wylaczone. UVLO 13,53 / 12,51 V (nominalnie).

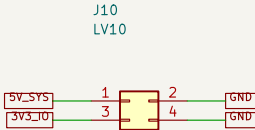
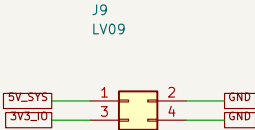
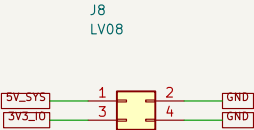
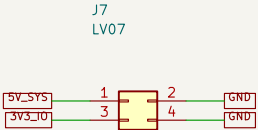
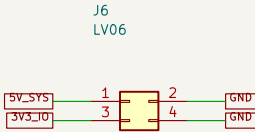
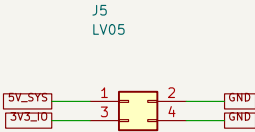
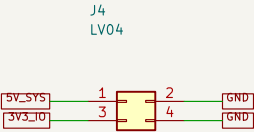
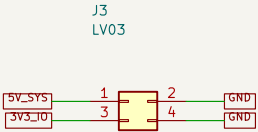
P04_3V3 (J13.1) to 3V3 wracajace z P04 – celowo NIE polaczone z lokalnym 3V3_IO (SAFE_N = L bez ENABLE).

Odstępstwo od D-06 (dosł.): U2A buforuje OK zamiast porównywac UV_CMP – PFAIL_N nie może rozjechać się ze stanem Q1.

VLOG -> F2/F3 1A MINI -> TSR 2-2450 (5V_SYS), TSR 2-2433 (3V3_I0) - jak R3



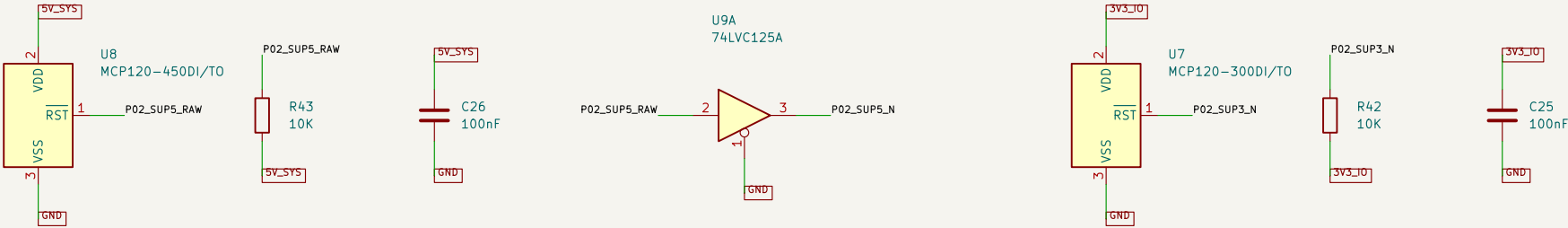
LV03..LV10: 1=5V_SYS 2=GND 3=3V3_I0 4=GND (bez zmian)



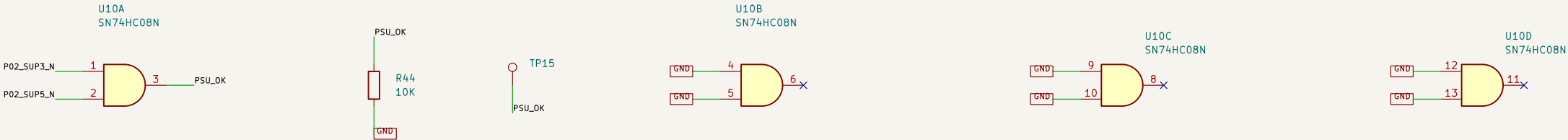
BUDZET LACZNY <= 6 W na VLOG (ze stratami przetwornic). F2/F3: MINI 1A 32 VDC (D-04).

Dodatkowa pojemnosc obciazenia: suma <= 600 uF na 5V i <= 900 uF na 3V3 (wlicz C22/C24).

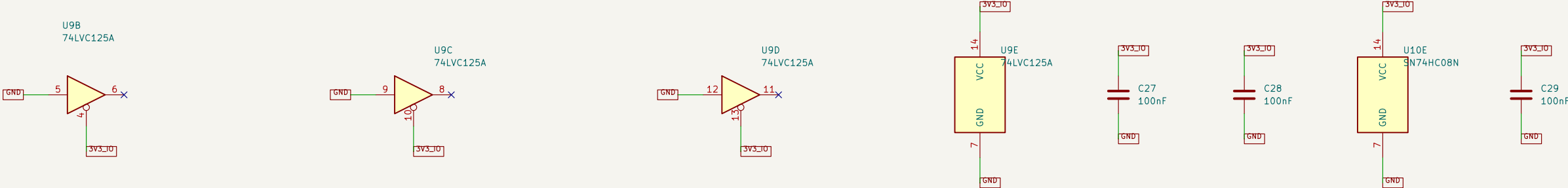
Nadzor szyn (jak R3): U8 5V_SYS, U7 3V3_I0, U9A przesuwą poziom



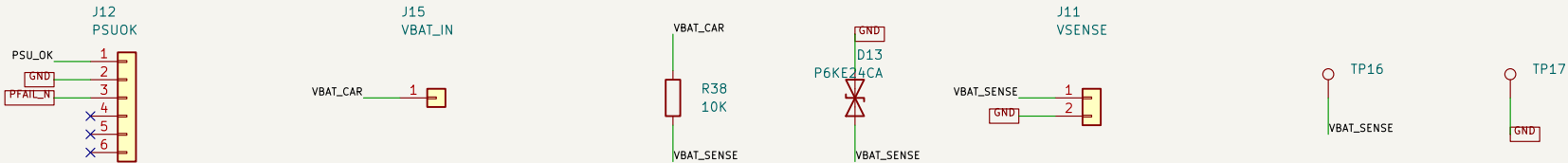
PSU_OK = SUP3_N & SUP5_N (U10A); bramki B-D nieuzywane, wejscia do GND



U9B-D nieuzywane (wejscia do GND, OE# do 3V3_I0); zasilanie U9/U10



J12 PSUOK do P04 (pin 3 = PFAIL_N); VBAT auta: J15 - R38 10k - D13 P6KE24CA - J11 VSENSE do P05 CH7



U9A: wejscie toleruje 5 V; zasilanie 3V3. C28 na adapterze przy samym IC (nie na plycie).

VBAT: jeden przewod od klemy + akumulatora (bezpiecznik 0,5 A przy klemie), bez przewodu masy (D-01).